American Biochar Institute

Biochar in Poultry Houses

Scientific Basis for Ammonia Control and Litter Value

Ammonia is one of the most persistent problems in poultry production. It forms when microbes break down the nitrogen in manure, and it builds up fastest in warm, moist bedding when houses are closed for heating. Elevated ammonia irritates bird eyes and airways, slows weight gain, raises fuel costs by forcing extra ventilation, and drives nitrogen losses that reduce the fertilizer value of the litter. Industry guidance generally targets ammonia concentrations below 25 parts per million (ppm) at bird level.

Biochar, the carbon-rich solid produced by heating biomass under low-oxygen conditions (pyrolysis), has physical and chemical properties suited to this problem: very high internal surface area, strong porosity, and an ability to bind ammonia and ammonium. This document summarizes what on-farm trials, controlled litter studies, and composting research say about using biochar in poultry houses. It covers three routes of use, **bedding amendment**, **feed supplement**, and **litter and compost additive**, and identifies where the evidence is strong, where it is emerging, and what remains unknown, so growers and technical advisors can match the practice to their goals.

1. Key Concepts

A few terms frame the evidence that follows:

- **Litter:** the mix of bedding material (pine shavings, rice hulls, miscanthus, or similar) and accumulated manure on the poultry house floor.
- **Ammonia volatilization:** the conversion of nitrogen in manure (mainly uric acid and urea) into ammonia gas that escapes into house air. Volatilized nitrogen is lost from the litter, so high ammonia means both worse air quality and a less valuable fertilizer product.
- **Adsorption:** the binding of a gas or dissolved compound onto a surface. Biochar's internal pore network gives a single gram of material a surface area of hundreds of square meters, which is what allows it to capture ammonia and ammonium.
- **Application routes:** biochar can be mixed into or spread on top of bedding, fed to birds as a diet supplement (after which it exits in the manure), or blended into litter during storage and composting. The three routes act at different points in the same nitrogen pathway.

The core logic is consistent across all three routes: capture nitrogen before it escapes as gas, keep it in the litter, and carry that value through to the field.

2. Evidence from Working Poultry Houses

The most direct evidence comes from trials inside commercial broiler houses.

A multi-flock, on-farm trial in Rockingham County, Virginia tracked ammonia levels across three broiler houses on one farm from 2023 through early 2026, with one house receiving biochar and the others serving as comparisons.[1] The trial evolved from single pre-flock applications (three supersacks pulverized into the bedding) to weekly surface additions of roughly 180 liters (ten 5-gallon buckets) of biochar. Results strengthened as the protocol shifted to frequent surface applications. In the May 2025 flock, the biochar house registered no detectable ammonia by day 37 even though the house still had normal animal odor, and in later flocks the biochar house consistently showed the lowest ammonia readings by mid-flock.[1]

Table 1 summarizes the arc of that trial. Two honest caveats apply. First, this is a single-farm trial without replication, and the three houses differ in size, ventilation design, and flock size, so weather and ventilation confound individual comparisons. Second, results early in each flock were inconsistent; the biochar advantage typically emerged after two to four weeks of additions.[1]

Flock start	Biochar treatment (House 3)	Observed outcome
Mar 2023	3 supersacks mixed into partially fresh bedding	Biochar house dropped below control after day 15
Jan 2025	None (baseline flock, identical fresh bedding)	No consistent advantage; moisture drove readings
May 2025	Weekly surface additions, ~180 L/week from day 8	No detectable ammonia at day 37; lowest at day 44
Aug 2025	Weekly additions continued, no cleanout	Biochar house lowest on nearly every reading
Feb 2026	Full cleanout; 12 to 16 buckets weekly	Highest house at day 4 (pre-biochar); clearly lowest after 4 additions

Table 1. Summary of the Rockingham County, Virginia on-farm broiler house trial (Teel, 2026). PPM values are averages of in-house readings.

A controlled study at a commercial scale points the same general direction on safety and litter quality. Linhoss and colleagues applied biochar to pine shavings at 0.97 kilograms per square meter (about 0.2 pounds per square foot) before flock placement and found no negative effects on broiler performance, health, or litter nutrient content through 35 days.[2] In the same study, blending biochar into shavings at 10 to 20 percent by weight increased the bedding's water holding capacity by 22 to 32 percent, which matters because wet litter is the precondition for both ammonia release and footpad lesions.[2] The Virginia trial, by contrast, saw no clear reduction in bedding moisture from biochar alone, so moisture control should be treated as a possible secondary benefit rather than a guaranteed one.[1]

3. Why Biochar Lowers Ammonia

Three mechanisms, documented mainly in litter and composting research, explain the field observations:

- **Physical adsorption.** Ammonia gas and dissolved ammonium bind to biochar's pore surfaces and oxygen-containing surface groups, holding nitrogen in the litter instead of the air.[3,4]
- **Microbial nitrogen incorporation.** Biochar appears to shift litter microbiology toward incorporating ammonium into organic compounds and, potentially, toward converting it to nitrate. Composting studies attribute part of the ammonia reduction to greater microbial use of nitrogen, not adsorption alone.[4]
- **Moisture buffering.** Where biochar raises the bedding's water holding capacity, it can keep the litter surface drier and slow the microbial reactions that generate ammonia in the first place.[2]

One caution: most biochars are alkaline, and raising litter pH can, in principle, push the ammonium-ammonia balance toward the gas phase. The trials above did not observe this in practice at the rates used, but it is a reason to start with moderate application rates and monitor, rather than assume more is better.[3,7]

4. Evidence from Litter Composting and Downstream Value

The largest body of quantitative evidence comes from composting poultry litter, which is where much litter ends up and where nitrogen losses are steepest. A systematic review of biochar in animal waste composting found that application rates of roughly 5 to 10 percent can extend the hot (thermophilic) composting phase, reduce leachate, and cut ammonia, methane, and nitrous oxide emissions.[3]

Individual poultry litter studies put numbers on this:

- Co-composting poultry litter with greenwaste or poultry litter biochar cut ammonia emissions by 55 to 60 percent relative to unamended compost, and raised nitrogen retention in the finished compost from 67 percent to as much as 84 percent of initial nitrogen.[4]
- Steiner and colleagues similarly reported reduced nitrogen loss when biochar was added to composting poultry litter.[5]
- Wood biochar at 5 and 10 percent reduced gaseous ammonia emissions from composting poultry manure by 30 and 44 percent, respectively.[6]
- Dose matters. At only 3 percent, biochar accelerated organic matter breakdown and nitrogen mineralization but had no relevant effect on gas emissions, a useful reminder that low rates may improve compost handling without delivering the emissions benefit.[7]
- Beyond ammonia, biochar-amended poultry litter compost emitted roughly one third the nitrous oxide of unamended compost, and total greenhouse gas emissions fell from 183 to 50 to 63 kilograms of CO_2-equivalent per tonne.[8]

Nitrogen kept in the litter is fertilizer value kept on the farm. Blending loblolly pine biochar into poultry litter at 10 to 20 percent improved nitrogen and phosphorus retention and the litter's physical handling properties.[9] For growers who sell litter, biochar treatment is therefore a reasonable candidate for increasing product value, though premium pricing for biochar-amended litter is not yet an established market and should not be assumed in budgets.

5. Feeding Biochar: The Dietary Route

Biochar can also enter the litter through the bird. A review of 112 studies on biochar in animal feeding found mostly positive or neutral effects across species, including toxin adsorption, digestion, and feed efficiency, and noted that biochar excreted in manure becomes enriched with nitrogen, making the manure a more valuable fertilizer with lower storage emissions.[10]

Poultry-specific feeding trials are more mixed and dose-sensitive:

- In broilers, beechwood biochar at 2 to 4 percent of the diet reduced ammonia emissions from manure by up to 17 percent, but the highest dose of a biochar blend worsened feed conversion by 8 percent and reduced final body weight by up to 7 percent.[11]
- In laying hens, low-rate supplementation (1 to 3 percent) improved laying performance by about 6 percent and eggshell strength by up to 10 percent, while ammonia reductions were positive in direction but not statistically significant.[12]
- A 2024 assessment concluded that dietary biochar's health benefits are best established precisely where a toxin such as ammonia is present in the shed atmosphere; where no such stressor exists, productivity gains are less likely.[13]

The Virginia on-farm trial adds a practical observation consistent with this route: biochar appeared more effective when surface-applied where birds could peck and consume it, suggesting the bedding and dietary pathways reinforce each other.[1] Growers under contract should confirm any feed additive with their integrator before use.

6. Practical Guidance for Growers

Table 2 consolidates the application rates that have been tested. These are starting points, not prescriptions; biochar properties vary widely with feedstock and production temperature, and none of these rates has been optimized across house types.[14]

Route	Tested rate	Evidence and notes
Pre-flock surface application	~1 kg/m² (0.2 lb/ft²) on bedding	No negative effects on performance, health, or litter nutrients[2]
Repeated surface application	~180 L (10 buckets) per house, weekly	Strongest in-house ammonia reductions in on-farm trial; benefit builds over 2 to 4 weeks[1]

Blended into bedding	10–20% by weight	Raised water holding capacity 22–32%; improved litter N and P retention[2,9]
Composting amendment	5–10% of mix	Reduced NH_3 30–60%, cut N_2O and total GHG; 3% too low for emission benefits[3,4,6,7,8]
Feed supplement	1–2% of diet (start low)	NH_3 down up to 17%; doses above ~4% risked performance losses; confirm with integrator[10,11,12]

Table 2. Tested biochar application rates for poultry systems. Superscripts refer to the numbered references.

Whatever the route, use biochar produced from clean feedstocks and characterized by the supplier (feedstock, production temperature, particle size, pH). Fine, dusty material applied in an enclosed house can create respirable dust; apply during low-ventilation moments, dampen slightly if needed, and wear a dust mask.

7. Executive Review

Established: Biochar reduces ammonia emissions and improves nitrogen retention when composted with poultry litter, with reductions of 30 to 60 percent at 5 to 10 percent inclusion documented across independent studies.[3,4,5,6,8] Surface application to bedding at about 1 kilogram per square meter is safe for bird performance and health.[2]

Emerging: Repeated in-house surface application appears to lower ammonia at bird level in commercial houses, with the strongest results after several weeks of weekly additions, but this rests on a single-farm trial with confounded comparisons and needs replication.[1] Dietary biochar at low rates can modestly reduce manure ammonia and may improve layer performance, with dose-dependent risks at higher inclusion.[11,12,13]

Unknown: The fate of the ammonia biochar captures, whether biochar assists the bacterial conversion of ammonia to nitrate in litter, its effect on nitrous oxide from litter after field application, and the optimal in-house application rate all remain open questions.[1,14] Farm-scale performance verification and standardized biochar specifications for this use are the field's most pressing needs.[14]

For a grower dealing with ammonia problems, the practical takeaway is straightforward: the composting evidence is solid, the in-house evidence is promising but early, and the entry cost of a trial (surface application in one house, with weekly ammonia readings against a comparison house) is low.

References

1. Teel, W.S. (2026). Biochar and Poultry: The Impact on Ammonia. Presentation, James Madison University (Professor Emeritus). On-farm broiler house trial data, Rockingham County, Virginia, 2023–2026.

2. Linhoss, J.E., Purswell, J.L., Street, J.T., & Rowland, M.R. (2019). Evaluation of Biochar as a Litter Amendment for Commercial Broiler Production. Journal of Applied Poultry Research, 28(4), 1089–1098.

3. Akdeniz, N. (2019). A systematic review of biochar use in animal waste composting. Waste Management, 88, 291–300.

4. Agyarko-Mintah, E., Cowie, A., Van Zwieten, L., Singh, B.P., Smillie, R., Harden, S., & Fornasier, F. (2017). Biochar lowers ammonia emission and improves nitrogen retention in poultry litter composting. Waste Management, 61, 129–137.

5. Steiner, C., Das, K.C., Melear, N., & Lakly, D. (2010). Reducing Nitrogen Loss during Poultry Litter Composting Using Biochar. Journal of Environmental Quality, 39(4), 1236–1242.

6. Janczak, D., Malińska, K., Czekała, W., Cáceres, R., Lewicki, A., & Dach, J. (2017). Biochar to reduce ammonia emissions in gaseous and liquid phase during composting of poultry manure with wheat straw. Waste Management, 66, 36–45.

7. Sánchez-García, M., Alburquerque, J.A., Sánchez-Monedero, M.A., Roig, A., & Cayuela, M.L. (2015). Biochar accelerates organic matter degradation and enhances N mineralisation during composting of poultry manure without a relevant impact on gas emissions. Bioresource Technology, 192, 272–279.

8. Agyarko-Mintah, E., Cowie, A., Singh, B.P., Joseph, S., Van Zwieten, L., Cowie, A., Harden, S., & Smillie, R. (2017). Biochar increases nitrogen retention and lowers greenhouse gas emissions when added to composting poultry litter. Waste Management, 61, 138–149.

9. Mohammadi-Aragh, M.K., Stokes, C.E., Street, J.T., & Linhoss, J.E. (2021). Effects of Loblolly Pine Biochar and Wood Vinegar on Poultry Litter Nutrients and Microbial Abundance. Animals, 11(8), 2209.

10. Schmidt, H.-P., Hagemann, N., Draper, K., & Kammann, C. (2019). The use of biochar in animal feeding. PeerJ, 7, e7373.

11. Kalus, K., Konkol, D., Korczyński, M., Koziel, J.A., & Opaliński, S. (2020). Effect of Biochar Diet Supplementation on Chicken Broilers Performance, NH3 and Odor Emissions and Meat Consumer Acceptance. Animals, 10(9), 1539.

12. Kalus, K., Konkol, D., Korczyński, M., Koziel, J.A., & Opaliński, S. (2020). Laying Hens Biochar Diet Supplementation – Effect on Performance, Excreta N Content, NH3 and VOCs Emissions, Egg Traits and Egg Consumers Acceptance. Agriculture, 10(6), 237.

13. Hegarty, R., Joseph, S., Rebbeck, M., Meale, S., & Paul, N. (2024). Biochar as an animal feed ingredient. In Biochar for Environmental Management (3rd ed.). Routledge.

14. Chen, B., Koziel, J.A., Bialowiec, A., & O'Brien, S.C. (2024). The potential role of biochar in mitigating gaseous emissions from livestock waste – A mini-review. Journal of Environmental Management, 370, 122692.